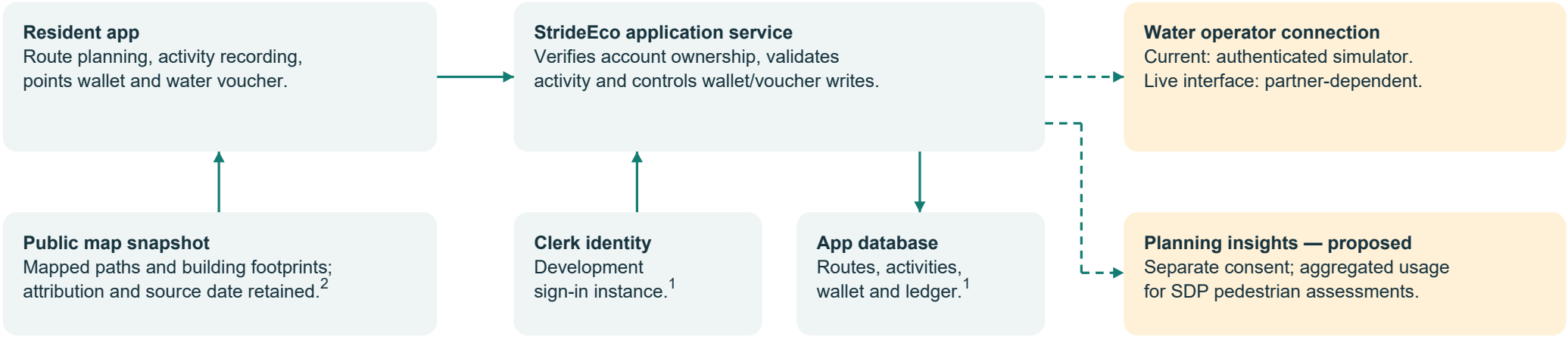
How the system fits together

PACE: Plan · Act · Collect · Exchange. Phone and desktop support planning and rewards; the phone is the intended activity recorder.[1]



Component	Recorded position on 23 October 2026	Boundary
Interface & identity	Responsive PACE screens and development sign-in have recorded checks.[1]	Production authentication configuration and full pilot readiness remain separate requirements.
Shared storage	Dedicated database provisioned; four migrations verified.[1]	Latest retained storage report leaves deployment activation and hosted persistence verification pending.
Water & insights	Machine-bound code/QR and accounting are implemented in the simulator.[1]	Mock locations are illustrative. Physical dispensing and resident planning insights are not established.

Diagram: solid connections describe the prototype; the dashed operator connection requires a compatible partner. Implementation status is based on retained project evidence, not a new live-service check.

Route planning, shade & activity data

Public geographic inputs support a bounded prototype. Estimated shade and recorded movement have distinct accuracy limits.

Function	Current design / evidence	Limit and required treatment
Public route network	OSM extract: 88 eligible walking ways and 582 building footprints around Elmina Valley Central Park / Elmina Urban Park.[2]	Keep source date and © OpenStreetMap contributors attribution. The extract is not a survey of path access, surface or safety.[2,5]
Route choice	Generate loop/out-and-back or destination routes over connected mapped paths; compare available alternatives.[1]	Requested distance is a target. Limited coverage or disconnected paths may produce no suitable route; show that result clearly.
Building-shadow scenario	Use building footprints, departure date/time and an explicit assumed height. The snapshot has no height tags or tree-canopy features.[2]	Shade is unavailable by default. Optional scenarios exclude tree shade, clouds and overhangs; projections are capped and unavailable at very low sun angles.[2]
Location recording	With permission, the foreground browser records time-stamped locations. Web geolocation requires a secure context and user permission.[1,3]	Device accuracy varies. Phone is intended for recording; desktop supports planning, saving, wallet and redemption. No dedicated sensor installation is required.
Background / screen-off	Foreground recording is the stated MVP mode. Browsers can throttle background work.[1,4]	Continuous screen-off tracking is not promised. Explain interruptions and test supported physical handsets before a resident pilot.
Credit eligibility	Server-side activity checks and replay-safe crediting are part of the prototype.[1]	Plausibility checks cannot authenticate device coordinates. Keep demo activity separately labelled and exclude it from real participation evidence.

Scope: direct browser location recording; a Strava integration is not required for this MVP. Points are distance-based. No shade bonus, measured cooling benefit or thermal-safety claim is included.

Points accounting & water redemption

Prototype rules below are team assumptions. The final earning rate, reward cost, funding and caps require agreement before a live pilot.[1]

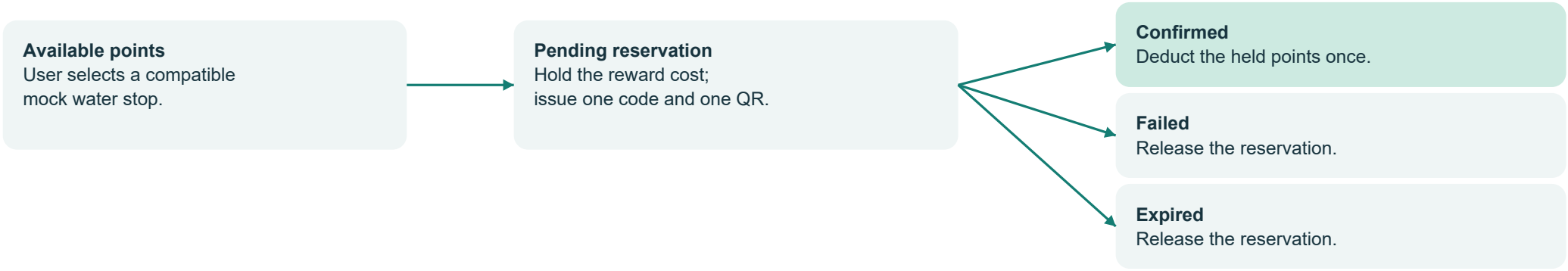
EARN
50 metres / point
Eligible distance only

REDEEM
60 points / water
Illustrative 500 ml reward

CAP
4 rewards / month
Pending + confirmed allocation

RESERVE
15 minutes
Prototype voucher lifetime

Prototype voucher states



Control	What it protects
One voucher, two entry methods	Manual code and QR represent the same single use. Downloading the QR creates no extra reward.[1]
Owner + machine checks	The authenticated server checks account, credential, assigned machine, type, pending state and expiry before simulated consumption.[1]
Atomic accounting	An atomic state change and ledger rules prevent duplicate spending in the simulator. A confirmed voucher cannot be reused through the other entry method.[1]

Live integration requirement: an operator timeout can leave dispensing uncertain. Add an awaiting-confirmation/reconciliation process before live use; do not automatically refund or issue another dispense while the physical outcome is unknown. See page 4.

What a live pilot needs

These are proposed launch requirements. They do not imply an existing machine inventory, operator agreement or funded water supply.

Dependency / owner	Minimum agreement or capability	Failure handling / launch check
Machine & water supply Operator + SDP/site team	Confirm an actual site, drinking-water product, compatible interface, stock and restocking responsibility.	Disable unavailable stops. Confirm that the proposed redemption method can dispense the agreed water reward.
Funding & settlement SDP/sponsor + operator	Agree who pays per confirmed water reward, programme budget, caps and settlement records.	Reconcile confirmed dispensing to vouchers and funding. Pause new rewards when stock or budget is unavailable.
Operator integration Technical team + operator	Authenticate the operator, bind requests to a machine and use stable transaction IDs with repeat-safe handling.	Test duplicates, delayed responses, wrong machines, failure and unknown outcomes. Reconcile uncertainty before release or retry.
Identity & persistence Technical team	Activate and verify hosted storage; configure authentication appropriate for the pilot. Production Clerk uses a separate production setup.[1,6]	Verify save/reload, cross-device wallet consistency and ownership isolation. Block account writes while data is unavailable.
Phone & route readiness Technical team + pilot lead	Check physical handset recording and explain foreground limits; confirm a bounded, suitable route area.	Test permission refusal, weak location and interruptions. Show uncertainty; mapped routes do not establish access or safety.

Proposed data controls

Separate choices: location permission for recording; separate consent for planning analytics. **Restricted access:** account-level ownership checks; SDP receives aggregated insights rather than identifiable movement trails. **Before launch:** agree retention, deletion, aggregation thresholds and access responsibilities; avoid exposing home/start locations in planning outputs.

Launch gate: compatible operator + confirmed water funding + verified hosted account flow + successful end-to-end dispensing and failure tests. Link the acceptance results to the roadmap; pilot KPI targets remain in A3.

Verification evidence & references

Evidence snapshot: retained project files inspected on 23 October 2026. This appendix adds no new live-deployment or physical-device test result.

Evidence level	What the retained records establish	What they leave open
Local / isolated tests	Recorded tests cover account ownership, credit replay, voucher binding, concurrent redemption and failure/expiry accounting.[1]	Live operator reliability, handset GPS accuracy and resident behaviour.
Hosted UI / identity	Records include responsive public UI, development sign-in and account-unavailable/retry checks.[1]	A complete hosted persistence journey after database activation.
Database provisioning	Dedicated database configuration and four migration checks are recorded.[1]	The latest retained storage record explicitly requires deployment activation and hosted verification.

Reference register

1. StrideEco implementation and retained test records
README.md; docs/product-brief.md; docs/water-vouchers.md; docs/account-loading.md; docs/remote-storage-verification.json; lib/pilot.ts and lib/vouchers.ts. Internal team evidence; distinguishes local, hosted and simulated results.

2. StrideEco public-data audit and route model
docs/data-sources.md; public/data/map-provenance.json; lib/geo.ts. OSM snapshot retrieved 22 October 2026. Bounded coverage, missing inputs and assumed-height scenarios.

3. MDN — Geolocation API
Secure-context and permission requirements; device-provided positions.
Official documentation

4. MDN — Page Visibility API
Hidden-page detection and browser background throttling.
Official documentation

5. OpenStreetMap — Copyright and licence
© OpenStreetMap contributors; ODbL attribution and data-licence requirements.
Licence / attribution

6. Clerk — Deploy to production
Production instance, credentials and deployment configuration requirements.
Official deployment guidance